Profit Leak Analysis Deep Analysis Conclusions

Some Important metrics:

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Revenue: 7998583.140000001
Product cost: 3931229.6899999995
Product margin: 4067353.4499999997
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1: Profitability baseline

	Metric	Amount
0	Revenue	7,998,583.14
1	Product Cost	-3,931,229.69
2	Refunds	-641,480.68
3	Return Shipping	-115,163.74
4	Restocking	-25,674.73
5	Outbound Shipping	-283,476.86
6	Contribution Profit	3,001,557.44

Conclusion:

The business generates about $8.0M in completed-order revenue, but after product costs, returns and outbound shipping, only about $3.0M remains as contribution profit.

2:Revenue and Profitability trend

year	Revenue	ProductCost	ProductMargin	Refunds	ReturnShipping	Restocking	OutboundShipping	ContributionProfit	ContributionMarginPct
2024	4,035,042.19	1,984,582.97	2,050,459.22	317,579.04	57,149.10	12,672.82	143,005.13	1,520,053.13	37.67
2025	3,963,540.95	1,946,646.72	2,016,894.23	323,901.64	58,014.64	13,001.91	140,471.73	1,481,504.31	37.38

Conclusion:

Revenue fell **1.77%** in 2025. Product margin percentage barely changed, but contribution profit fell faster, showing that downstream costs are important to the profitability story. Contribution Margin declined by **-2.54%.**

3: Category economics

	category	Revenue	ProductCost	Units	Orders	ProductMargin	MarginPct
1	Apparel	1,683,707.90	951,663.62	39041	20338	732,044.28	43.48
5	Home & Kitchen	1,482,017.26	610,884.57	32202	17281	871,132.69	58.78
3	Electronics	1,346,858.87	439,246.34	30230	16258	907,612.53	67.39
0	Accessories	1,164,637.30	655,755.53	23163	12611	508,881.77	43.69
2	Beauty	921,443.22	538,926.30	21694	11911	382,516.92	41.51
4	Footwear	751,762.66	401,530.40	17322	9567	350,232.26	46.59
6	Sports	648,155.93	333,222.93	15840	8739	314,933.00	48.59

Conclusion:

Apparel is the largest revenue category but also carries the highest return burden. Electronics and Home & Kitchen retain considerably more value.

4: Profit leakage candidates

category	Revenue	ProductCost	ProductMargin	MarginPct	Units	Orders	ReturnedUnits	ReturnRatePct
Apparel	1,683,707.90	951,663.62	732,044.28	43.48	39041	20338	4401	11.27
Footwear	751,762.66	401,530.40	350,232.26	46.59	17322	9567	1791	10.34
Accessories	1,164,637.30	655,755.53	508,881.77	43.69	23163	12611	1739	7.51
Electronics	1,346,858.87	439,246.34	907,612.53	67.39	30230	16258	2088	6.91
Home & Kitchen	1,482,017.26	610,884.57	871,132.69	58.78	32202	17281	2154	6.69
Sports	648,155.93	333,222.93	314,933.00	48.59	15840	8739	1053	6.65
Beauty	921,443.22	538,926.30	382,516.92	41.51	21694	11911	1284	5.92

Conclusion:

Apparel is the clearest high-scale/high-return category and should be investigated first.

5:Product Economics

	product_id	category	Units	Revenue	MarginPct	ReturnedUnits	ReturnRatePct	TotalReturnCost	ContributionBeforeOutboundShipping
457	P00458	Footwear	225	49,932.88	52.06	27	12.00	6,412.51	19,581.09
229	P00230	Apparel	181	37,499.49	54.10	24	13.26	5,318.32	14,968.85
935	P00936	Footwear	207	39,603.60	49.70	20	9.66	4,049.13	15,635.40
395	P00396	Apparel	227	31,048.57	41.35	27	11.89	4,035.38	8,802.73
700	P00701	Apparel	181	24,331.29	35.84	27	14.92	3,939.91	4,779.21
382	P00383	Apparel	233	25,206.78	34.98	31	13.30	3,627.60	5,190.55
279	P00280	Electronics	200	26,937.70	75.27	22	11.00	3,271.20	17,004.02
674	P00675	Home & Kitchen	189	33,649.18	66.87	17	8.99	3,250.10	19,250.38
907	P00908	Accessories	200	33,834.49	56.49	16	8.00	2,917.41	16,195.90
665	P00666	Apparel	209	22,073.45	30.59	24	11.48	2,851.68	3,900.04
267	P00268	Apparel	187	22,027.46	37.45	22	11.76	2,810.98	5,438.74
265	P00266	Footwear	205	15,062.03	46.16	34	16.59	2,795.39	4,156.78
75	P00076	Apparel	178	20,989.13	46.26	21	11.80	2,755.43	6,954.91
352	P00353	Apparel	194	24,784.34	31.29	20	10.31	2,751.55	5,003.34

Conclusion:

Several high-volume products show unusually high return rates. These products deserve product-level investigation rather than judging the entire category alone.

6:Discount Economics

discount_rate	Revenue	ProductCost	Units	Orders	ProductMargin	MarginPct
0.00	4,075,121.04	1,898,874.39	86120	35029	2,176,246.65	53.40
0.05	1,622,146.31	787,868.18	35859	14402	834,278.13	51.43
0.10	1,081,002.74	554,527.17	25433	10359	526,475.57	48.70
0.15	652,398.51	355,011.81	16382	6518	297,386.70	45.58
0.20	375,886.53	216,998.37	10231	4264	158,888.16	42.27
0.25	192,028.01	117,949.77	5467	2200	74,078.24	38.58

Conclusion:

Higher discounts are associated with progressively weaker product margins. The margin difference between no discount and a 25% discount is substantial.

7:Discount and Returns

discount_rate	Items	ReturnedItems	PurchasedUnits	ReturnedUnits	ReturnedItemRatePct
0.00	48932	6879	86120	6,932.00	14.06
0.05	20203	2926	35859	2,950.00	14.48
0.10	14489	2092	25433	2,105.00	14.44
0.15	9224	1261	16382	1,273.00	13.67
0.20	5947	832	10231	836.00	13.99
0.25	3092	412	5467	414.00	13.32

Conclusion:

Higher discounts do not show a clear increase in return behavior. The stronger observed issue is margin erosion.

8:Coupon Economics

CouponUsed	Orders	Revenue	ProductCost	Units	ProductMargin	MarginPct
False	36557	4,236,223.78	1,981,444.94	89911	2,254,778.84	53.23
True	36215	3,762,359.36	1,949,784.75	89581	1,812,574.61	48.18

Conclusion:

Coupon orders have lower observed margin than non-coupon orders, suggesting that discount-driven revenue should be evaluated against the value actually retained.

9:Return Reasons

	return_reason	Events	ReturnedUnits	Refunds	ReturnShipping	Restocking	TotalReturnCost	CostPerReturnedUnit
0	Changed Mind	3050	3056	135,105.88	24,446.89	5,390.88	164,943.65	53.97
3	Not as Expected	2813	2815	125,583.25	22,185.39	4,951.21	152,719.85	54.25
2	Late Delivery	2140	2147	94,664.85	16,944.14	3,807.45	115,416.44	53.76
6	Wrong Size	1986	1995	84,386.70	15,855.56	3,406.75	103,649.01	51.95
1	Damaged in Transit	1619	1621	73,249.44	12,842.38	2,928.86	89,020.68	54.92
4	Product Defect	1450	1455	64,713.25	11,675.68	2,610.71	78,999.64	54.30
5	Wrong Item	1421	1421	63,777.31	11,213.70	2,578.87	77,569.88	54.59

Conclusion:

Changed Mind and Not as Expected are the biggest return-cost drivers, together representing about 41% of return-related cost.

10:Fulfillment

	delivery_status	Shipments	ShippingCost	AvgShippingCost	AvgDeliveryDays
0	Late	11740	114,153.94	9.72	3.88
1	On Time	18260	169,322.92	9.27	2.14

Conclusion:

Late delivery is common, with roughly 39% of tracked shipments delivered late.

11:Late delivery vs Returns

delivery_status	Shipments	ReturnedOrders	ReturnRatePct
Late	11740	2203	18.76
On Time	18260	3443	18.86

Conclusion:

Late shipments do not have a meaningfully higher overall return rate than on-time shipments. Late delivery is therefore a service issue, but not a proven broad return driver But their is an interesting point also and that is: “Some customers explicitly return because of late delivery **(2140 returns** because of late delivery), but late shipments overall are not associated with a higher total return rate.”

12:Warehouse

	warehouse_id	Shipments	ShippingCost	AvgShippingCost	LateRate	ReturnRate	LateRatePct	ReturnRatePct
2	WH03	3963	37,297.66	9.41	0.40	0.19	39.62	18.72
5	WH06	3625	34,508.11	9.52	0.40	0.19	39.56	19.20
3	WH04	3153	29,605.43	9.39	0.40	0.18	39.55	18.46
6	WH07	3960	37,609.74	9.50	0.39	0.19	39.34	18.86
0	WH01	3635	34,222.87	9.41	0.39	0.19	39.34	18.73
1	WH02	3879	36,961.20	9.53	0.39	0.18	38.98	17.79
7	WH08	3670	34,454.39	9.39	0.38	0.19	38.45	19.21
4	WH05	4115	38,817.46	9.43	0.38	0.20	38.35	19.51

Conclusion:

No warehouse stands out dramatically. The fulfillment issue appears broader than one warehouse.

13:Customer Economics

CustomerSegment	Customers	Orders	Revenue	ProductMargin	Refunds	ReturnShipping	Restocking	Contribution
Highly Repeat	4066	25533	3,196,326.22	1,627,531.93	221,459.31	40,036.46	8,914.61	1,357,121.55
No Completed Orders	4178	0	0.00	0.00	0.00	0.00	0.00	0.00
One-Time	7244	7244	689,925.42	351,875.07	66,214.40	11,746.98	2,570.01	271,343.68
Repeat	14512	39995	4,112,331.50	2,087,946.45	353,806.97	63,380.30	14,190.11	1,656,569.07

Conclusion:

Repeat customers are substantially more valuable than one-time customers, with highly repeat customers producing the strongest contribution per customer.

14:Customer type Economics

customer_type	Customers	Orders	Revenue	ProductMargin	Contribution	Refunds	RevenuePerCustomer	ContributionPerCustomer
Consumer	26296	58721	5,030,459.47	2,556,812.50	1,927,277.32	516,324.10	191.30	73.29
Small Business	3704	14051	2,968,123.67	1,510,540.95	1,357,756.98	125,156.58	801.33	366.57

Conclusion:

Small Business customers are much more valuable per customer than consumers and contribute disproportionately relative to their customer-base size.

15:Loyalty

	loyalty_tier	Customers	Orders	Revenue	ProductMargin	Contribution	Refunds	RevenuePerCustomer	ContributionPerCustomer
0	Bronze	17976	20352	1,969,666.88	1,002,889.70	781,199.81	182,007.08	109.57	43.46
1	Gold	2056	14008	1,729,486.17	878,935.33	730,880.52	121,109.33	841.19	355.49
2	Platinum	234	2645	466,513.39	235,786.60	206,714.92	23,901.22	1,993.65	883.40
3	Silver	9734	35767	3,832,916.70	1,949,741.82	1,566,239.05	314,463.05	393.77	160.90

Conclusion:

Higher loyalty tiers are strongly associated with higher customer value. Platinum customers generate the highest contribution per customer.

16: Acquisition channel

acquisition_channel	Customers	Orders	Revenue	Contribution	RevenuePerCustomer	ContributionPerCustomer
Affiliate	3011	6041	663,677.36	268,162.73	220.42	89.06
Direct	4820	13259	1,467,324.55	601,591.91	304.42	124.81
Email	4191	11656	1,286,071.09	536,193.97	306.86	127.94
Google Ads	6662	14966	1,635,243.87	668,176.91	245.46	100.30
Meta	5968	12960	1,422,778.41	587,111.37	238.40	98.38
Organic	5348	13890	1,523,487.86	623,797.41	284.87	116.64

Conclusion:

Email-acquired customers have the highest contribution per customer, while Affiliate is lowest, but this does not account for acquisition cost.

17:Marketing Channel

channel	Spend	AttributedRevenue	NewCustomers	AttributedOrders	Impressions
Affiliate	4,679,496.20	15,838,161.50	92290	173565	346781311
Display	2,795,257.39	4,881,430.86	28644	53463	313420607
Email	4,025,337.87	82,763,197.95	492089	917964	346845292
Google Ads	10,271,276.46	36,917,634.32	215178	404637	389145564
Meta	4,238,231.25	20,373,945.18	120701	226032	312877555

Clicks	AttributedROAS	CostPerNewCustomer	CTR
5291664	3.38	50.70	1.53
2924098	1.75	97.59	0.93
2394679	20.56	8.18	3.57
8822698	3.59	47.73	2.27
5786230	4.81	35.11	1.85

Conclusion:

Email has the strongest attributed ROAS, while Display is weakest. These are attribution-based results and should not be treated as incremental returns.

18:Campaign type/objective

Campaign objectives show materially different attributed economics, with retention/repeat-purchase activity showing particularly high attributed ROAS.

19:Opportunity sizing

	return_reason	Events	ReturnedUnits	Refunds	ReturnShipping	Restocking	TotalReturnCost
0	Changed Mind	3050	3056	135,105.88	24,446.89	5,390.88	164,943.65
3	Not as Expected	2813	2815	125,583.25	22,185.39	4,951.21	152,719.85
2	Late Delivery	2140	2147	94,664.85	16,944.14	3,807.45	115,416.44
6	Wrong Size	1986	1995	84,386.70	15,855.56	3,406.75	103,649.01
1	Damaged in Transit	1619	1621	73,249.44	12,842.38	2,928.86	89,020.68
4	Product Defect	1450	1455	64,713.25	11,675.68	2,610.71	78,999.64
5	Wrong Item	1421	1421	63,777.31	11,213.70	2,578.87	77,569.88

CostPerReturnedUnit	PotentialSavings_10Pct	PotentialSavings_15Pct	PotentialSavings_20Pct
53.97	16,494.37	24,741.55	32,988.73
54.25	15,271.99	22,907.98	30,543.97
53.76	11,541.64	17,312.47	23,083.29
51.95	10,364.90	15,547.35	20,729.80
54.92	8,902.07	13,353.10	17,804.14
54.30	7,899.96	11,849.95	15,799.93
54.59	7,756.99	11,635.48	15,513.98

Conclusion:

A 10–20% reduction in return-related costs would represent roughly $78K–$156K in modeled potential savings.

20:Sales Channel

sales_channel	Revenue	ProductCost	ProductMargin	Orders	Units	ReturnedUnits	Refunds	ReturnShipping
Website	3,424,930.17	1,682,681.59	1,742,248.58	31289	76789	6204	276,673.18	49,383.49
Physical Store	1,988,528.10	978,583.45	1,009,944.65	18145	44879	3625	159,238.96	28,837.85
Mobile App	1,799,328.40	884,405.06	914,923.34	16171	40226	3236	141,924.39	25,492.81
Marketplace	785,796.47	385,559.59	400,236.88	7167	17598	1445	63,644.15	11,449.59

Restocking	TotalReturnCost	ContributionBeforeOutboundShipping	ContributionMarginPct	ReturnRatePct
11,041.78	337,098.45	1,405,150.13	41.03	8.08
6,407.81	194,484.62	815,460.03	41.01	8.08
5,725.01	173,142.21	741,781.13	41.23	8.04
2,500.13	77,593.87	322,643.01	41.06	8.21

Conclusion:

Sales channels have very similar contribution margins after return costs. No channel appears to be a major profitability leak.

21:Contribution by Year

year	Revenue	ProductCost	ProductMargin	Refunds	ReturnShipping	Restocking	OutboundShipping
2024	4,035,042.19	1,984,582.97	2,050,459.22	317,579.04	57,149.10	12,672.82	143,005.13
2025	3,963,540.95	1,946,646.72	2,016,894.23	323,901.64	58,014.64	13,001.91	140,471.73

ContributionProfit	ContributionMarginPct	ContributionProfitGrowthPct
1,520,053.13	37.67	NaN
1,481,504.31	37.38	-2.54

Conclusion:

Contribution profitability deteriorated slightly in 2025 even though product margin percentage remained stable. This strengthens the case that profitability leakage sits downstream of product margin.

5 strongest findings in the entire project

1. The company generates $8.0M revenue but only $3.0M contribution profit.

37.53% contribution margin after modeled direct costs.

2. Apparel is the clearest profitability pressure point.

$1.68M revenue + 11.27% return rate + $232.5K return cost.

3. Discounting is eating margin.

53.40% margin at 0% discount → 38.58% at 25%.

4. Repeat and Small Business customers are disproportionately valuable.

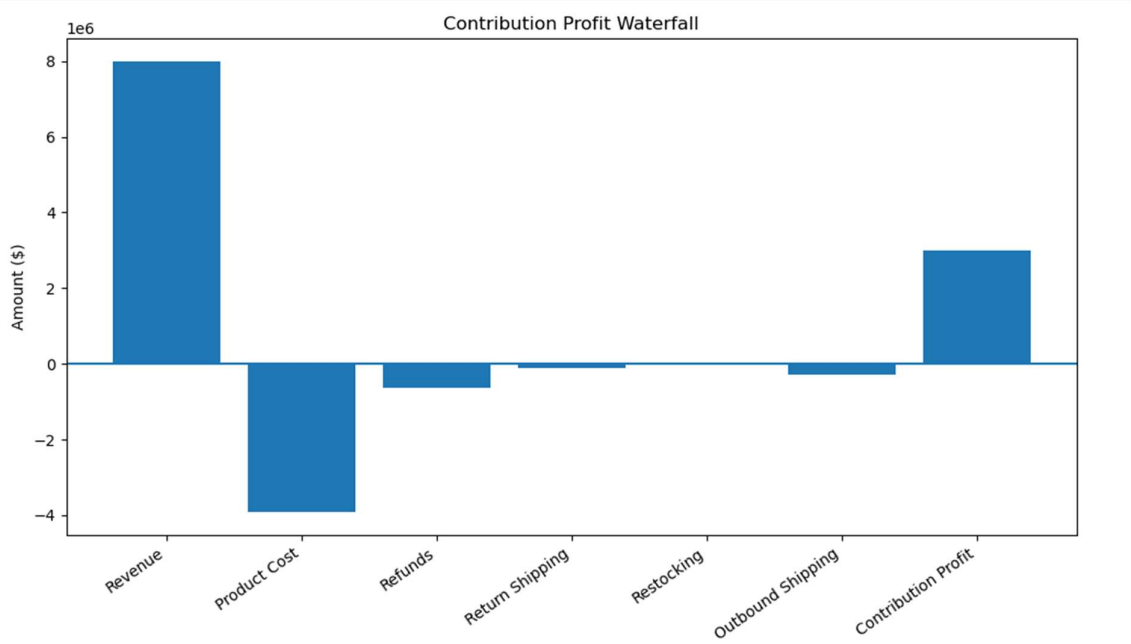
Highly repeat customers generate **$333.77 contribution/customer**, while Small Business generates **$366.57**.

5. Return costs are a meaningful profit leak.

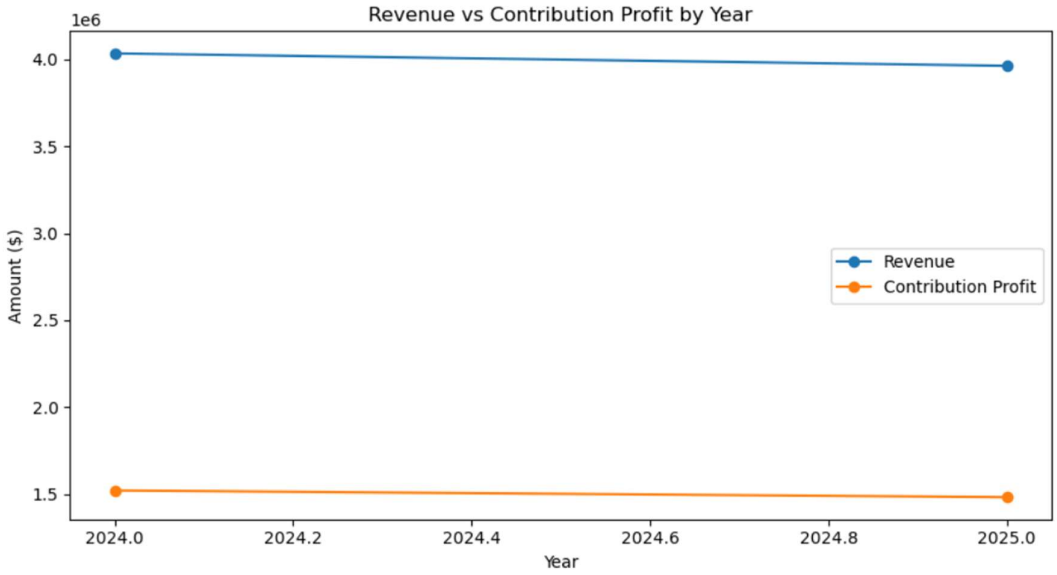
$782.3K of validated return-related costs, equal to about **9.78% of revenue** and **19.23% of product margin**.

Some Important Visuals

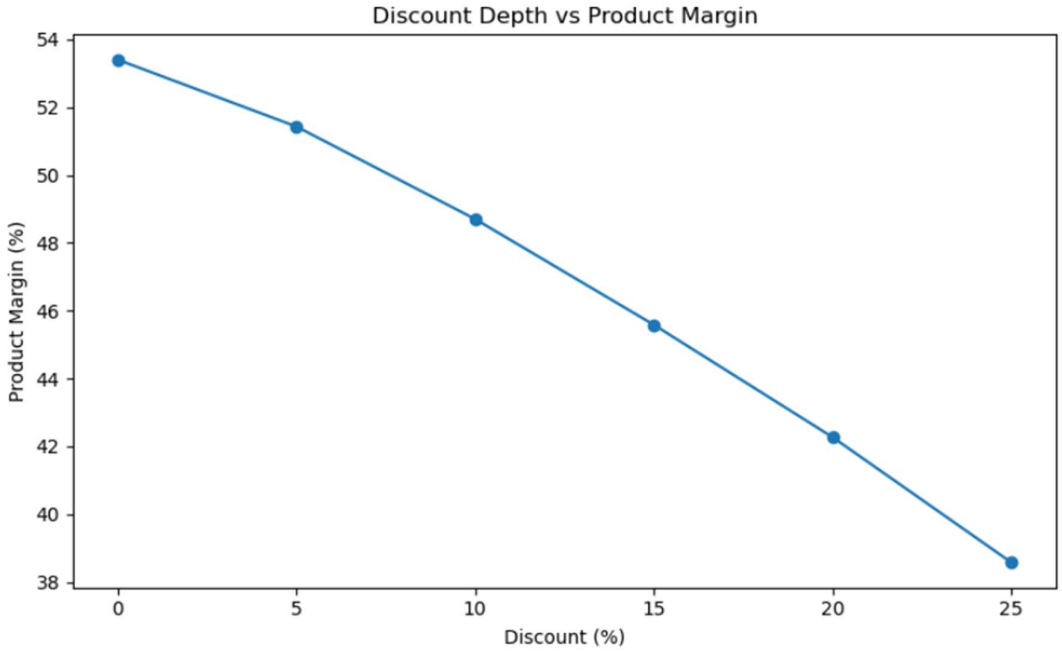
Contribution Profit Waterfall



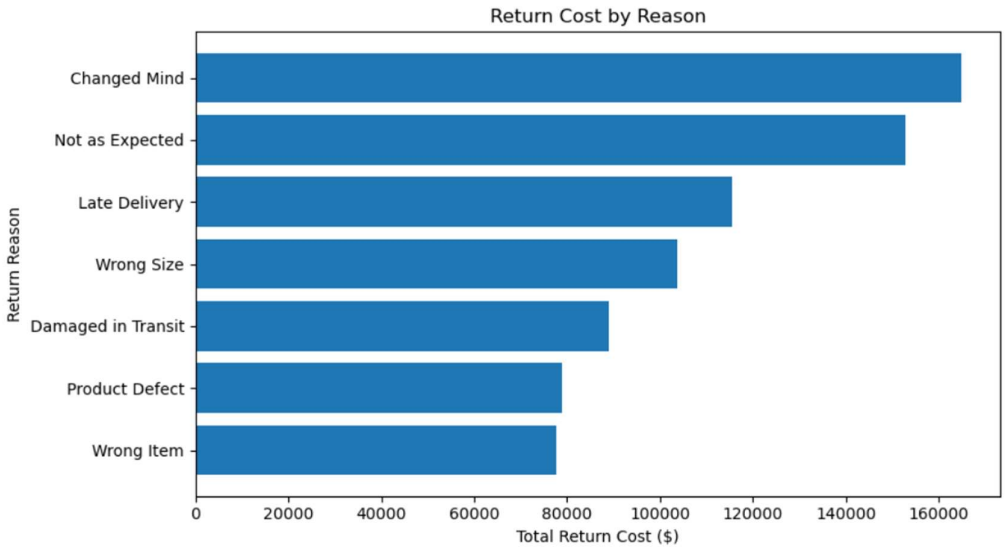
Revenue vs Contribution Profit



Discount vs Margin



Return Cost by reason



Final Business Strategy

Priority 1 — Recover profitable sales volume

Revenue fell 1.77%, so regain volume without relying heavily on discounts.

Priority 2 — Attack return leakage

Start with Apparel and the largest return-cost reasons.

Priority 3 — Protect margin

Deeper discounts are associated with substantial margin erosion.

Priority 4 — Grow valuable customers

Small Business and repeat customers generate substantially stronger economics.